

A single particle model with chemical/mechanical degradation physics for lithium ion battery State of Health (SOH) estimation

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HIGHLIGHTS

- Developed a single particle model coupled chemical/mechanical degradation physics.
- Capacity fade and voltage changes are accurately and quickly predicted during cycling.
- The impact of operating parameters on battery capacity and power are revealed.

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ABSTRACT

State of Health (SOH) estimation of lithium ion batteries is critical for Battery Management Systems (BMSs) in Electric Vehicles (EVs). Many estimation techniques utilize a battery model; however, the model must have high accuracy and high computational efficiency. Conventional electrochemical full-order models can accurately capture battery states, but they are too complex and computationally expensive to be used in a BMS. A Single Particle (SP) model is a good alternative that addresses this issue; however, existing SP models do not consider degradation physics. In this work, an SP-based degradation model is developed by including Solid Electrolyte Interface (SEI) layer formation, coupled with crack propagation due to the stress generated by the volume expansion of the particles in the active materials. A model of lithium ion loss from SEI layer formation is integrated with an advanced SP model that includes electrolytic physics. This low-order model quickly predicts capacity fade and voltage profile changes as a function of cycle number and temperature with high accuracy, allowing for the use of online estimation techniques. Lithium ion loss due to SEI layer formation, increase in battery resistance, and changes in the electrodes' open circuit potential operating windows are examined to account for capacity fade and power loss. In addition to the low-order implementation to facilitate on-line estimation, the model proposed in this paper provides quantitative information regarding SEI layer formation and crack propagation, as well as the resulting battery capacity fade and power dissipation, which are essential for SOH estimation in a BMS.

1. Introduction

Lithium Ion Batteries (LIBs) are key energy storage devices for many applications due to their high energy and power densities, and are widely used in Electric Vehicles (EVs) and Hybrid Electric Vehicles (HEVs). A critical challenge, however, is the capacity degradation experienced during repeated charge/discharge cycles. Battery performance declines over time due to irreversible physical and chemical changes that naturally occur until the battery can no longer be used. Therefore, State of Health (SOH) estimation is an essential component of a Battery Management System (BMS) for a variety of energy storage systems in transportation and stationary applications [1,2]. However,

there are challenges in performing SOH estimation in a BMS. First, SOH cannot be directly measured. Here, SOH refers to the state of a battery's condition compared to its initial condition, and is expressed as a loss of capacity relative to the initial value. For instance, when the battery capacity in EVs/HEVs reaches 80% of its initial capacity, the battery is no longer considered usable [1].

Secondly, battery performance continuously degrades due to various mechanisms, both mechanical and chemical, which affect LIBs during their lifetime. Therefore, studying these mechanisms require long-term, in situ testing that typically requires the battery to be destroyed. Moreover, it is challenging to decouple the effects of each of these mechanisms on battery health and performance. Modern LIBs,

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Nomenclature

a	crack length (m)	l_{cro}	initial crack width (m)
a_0	initial crack length (m)	L_{SEI}^0	initial SEI layer thickness (m)
$brug$	Bruggeman coefficient	L_i	electrode thickness (m)
c_0	initial concentration (mol m^{-3})	M_{SEI}	molecular weight of compounds constituting SEI (g mol^{-1})
$C_{\max, pos}$	positive maximum concentration (mol m^{-3})	n_{Li}	initial total number of lithium ions inside battery
$C_{\max, neg}$	negative maximum concentration (mol m^{-3})	n_{SEI}	consumed lithium ion for 1 mol of SEI layer formation
D_e	diffusion coefficient in electrolyte ($\text{m}^2 \text{s}^{-1}$)	N	cycle number
D_{sj}	solid-phase Li diffusivity ($\text{m}^2 \text{s}^{-1}$)	Q_0	battery capacity after formation cycle
E_{a1}	activation energy for crack propagation (kcal mol^{-1})	Q_{in}	battery capacity before SEI layer formation cycle
E_{a2}	activation energy for SEI layer growth (kcal mol^{-1})	Q_g	graphite specific capacity (Ah g^{-1})
E	Young's modulus of electrode material (N m^{-2})	Q_N	capacity after Nth cycle
F	Faraday's constant (C mol^{-1})	R	universal gas constant ($\text{J mol}^{-1} \text{K}^{-1}$)
$i = p/s/n$	positive/separator/negative	r_j	particle radius (m)
I_{app}	applied current density (A m^{-2})	R_{SEI}^0	initial SEI layer resistance (Ω)
$j = p/n$	positive/negative	t_+	cationic transport number
J_j^{Li}	molar flux density (mol s m^{-2})	$V(0)_N$	initial voltage at each cycle (V)
k_i	electrolyte conductivities (S m^{-1})	ε_i	electrode porosity
k_j	reaction rate constant ($\text{m}^{2.5} \text{mol}^{-0.5} \text{s}^{-1}$)	ρ_{cr}	number of cracks per unit area of particle (m^{-2})
k_0	crack propagation coefficient	ρ_g	graphite density (g cm^{-3})
K_{SEI0}	SEI layer growth coefficient	ρ_{SEI}	density of SEI films (g m^{-3})
k_{SEI}	SEI layer conductivity (S m^{-1})	Ω	partial molar volume of solute ($\text{m}^3 \text{mol}^{-1}$)

which are fabricated from a variety of anode and cathode materials, degrade due to a number of mechanisms that depend on the chemical nature of their constituent materials. In general, battery degradation mechanisms [3–6] include current collector corrosion, morphological changes of active materials, electrolyte decomposition, Solid Electrolyte Interphase (SEI) layer formation, and material dissolution. For example, carbonaceous materials, which are the most common anode materials in modern LIBs, have a significant amount of irreversible capacity loss during initial cycling as the SEI layer is formed on the carbon surface [3–5,7]. Moreover, the SEI layer continues to grow due to the continuous reduction of the electrolyte and the reformation of the SEI layers, which re-consumes lithium ions and results in irreversible battery capacity loss. For instance, about 8–15% irreversible capacity loss due to lithium ion loss is expected for MesoCarbon MicroBeads [5]. Further, mechanical damage to the battery will accelerate chemical degradation. The basic function of an electrochemical material is fulfilled by ions entering the active materials due to an electrochemical potential gradient [8]. During this intercalation process, volume change causes considerable stresses inside the particles, leading to mechanical failures such as pulverization of, or cracks and fractures in, the active materials. These cracks generate new surfaces on the particle, which are then exposed to the electrolyte, leading to additional SEI layer formation and the acceleration of capacity fade and power loss [8–14].

Despite these difficulties, the foundation of many models of different battery aging mechanisms has been presented by the research community to predict battery life. The studies in [15–20] utilized physics-based models that provide very detailed information regarding the battery electrochemical response. Capacity loss due to SEI layer formation was simulated through a continuum-scale mathematical model by considering the flux of the side reaction at the anode particle surface [15,16] or by modifying the solid phase concentration to be a function of the cycle according to lithium ion loss [17–19]. The effect of mechanical degradation on capacity fade has also been simulated using a full order physics-based model [20]. These models solve governing physical equations, which include mass conservation and charge conservation in the solid and electrolyte phases, as well as kinetic reactions at the interface between the solid and electrolyte.

A comprehensive and detailed state-of-the-art review of SOH estimation for LIBs has been conducted in [21]. These methods can be categorized as experimental techniques and adaptive methods.

Experimental techniques depend on recorded experimental data and previous knowledge about the effect of the operating conditions such as temperature, cycle number, SOC, current magnitude on the battery cycle life [21]. Although these methods are easy to implement onboard BMSs, their validity is limited to the calibration data used in their development. Furthermore, the wide range of operating conditions encountered in different battery applications, such as EVs, necessitates the use of an adaptive SOH estimation methods. Adaptive methods utilize parameters from battery models to estimate the SOH. Electrochemical models can provide highly accurate predictions of the battery behavior while providing insight into internal battery phenomena. Due to these advantages, these models are gaining popularity in recent years in the development of different BMS functionalities, such as SOH estimation. For BMSs, however, there is still a challenge in incorporating battery physics and degradation during on-line estimation.

Although a high fidelity electrochemical modes are ideal for the detailed analysis of battery phenomena, they are too computationally intensive to be efficiently utilized in a BMS [22]. This has led to efforts to reduce the complexity of electrochemical models, such as the studies in [23,24] that reduced the electrochemical model proposed by Doyle et al. [25] into a form suitable for a BMS. To further simplify battery models, most studies employ an Equivalent Circuit Model (ECM), which describes the battery dynamic behavior as a voltage source and a series of resistors and capacitors [26] and is widely used in BMSs due to the model's low complexity and ease of online implementation. However, ECM models have no physical significance, which leads to low fidelity and limited prediction capability. In addition, the prediction of battery side-reactions is not feasible due to the difficulty of obtaining the battery internal dynamic characteristics. Therefore, higher accuracies can be attained only by considering time-variant model parameters. The Single Particle (SP) model is a common type of reduced-order model. The SP model strikes the necessary balance between full order electrochemical models and ECMs, and is becoming a popular model in recent years for SOC and SOH estimation [27–31]. It assumes both electrodes are composed of multiple uniform sized spherical particles, and that the current distribution is uniform across both electrodes. Thus, each electrode can be approximated by a single spherical particle. The SP model is described by a set of ordinary differential equations, yet it is directly derived from comprehensive electrochemical models and, thus, explicitly retains many important battery properties with

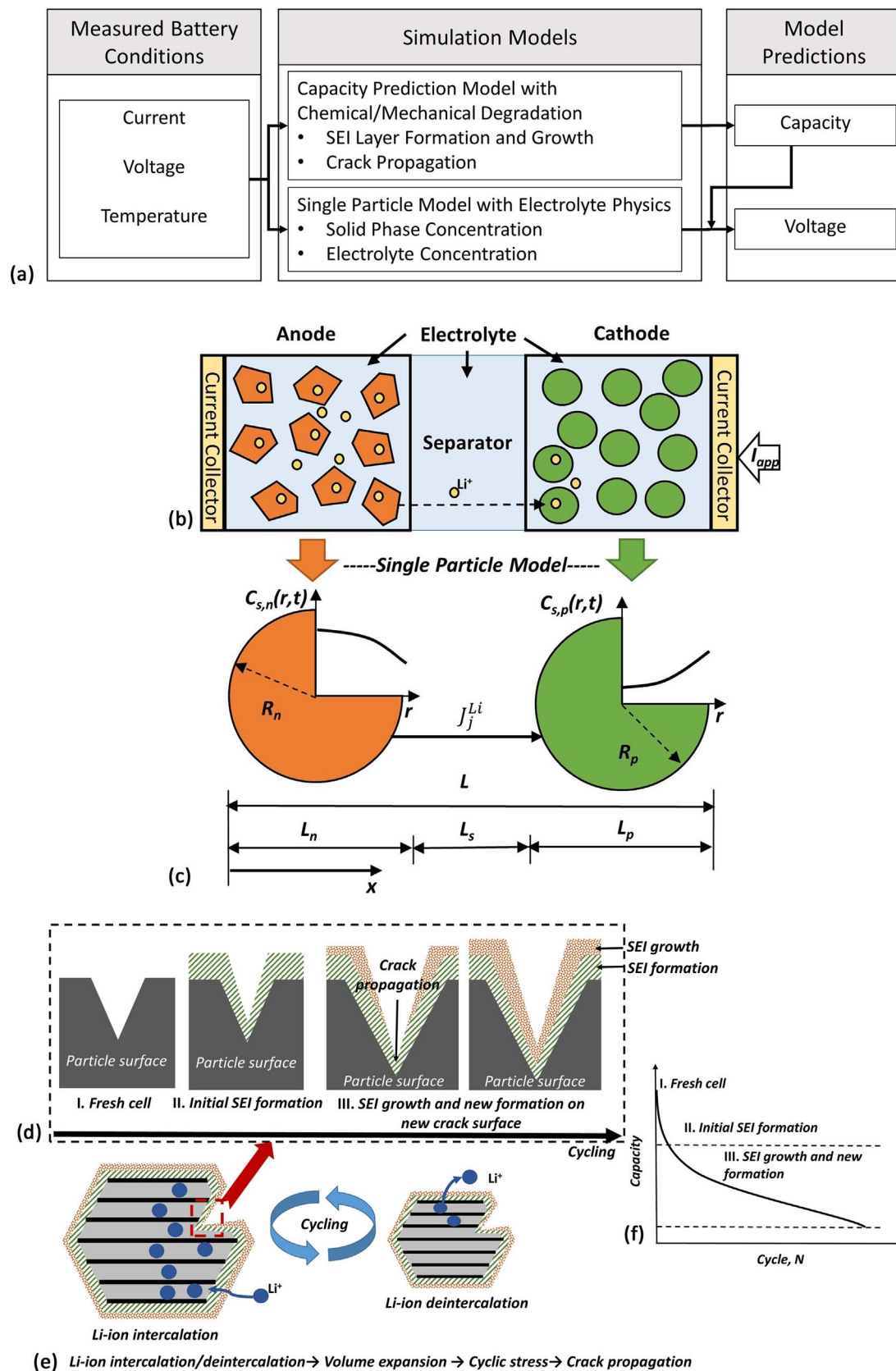


Fig. 1. Schematic diagrams for (a) overview of the proposed simulation processes, (b) a lithium ion battery composed of an anode, a separator, and a cathode, (c) representing two single particles for each electrode in the SP model, (d and e) coupled degradation mechanism between SEI layer formation and crack propagation on particle, and (f) capacity fade due to the lithium loss caused by SEI layer formation and growth.

high computational efficiency. The drawbacks of the SP model are that the accuracy of the SP model suffers at high C-rates due to the lack of electrolyte physics and degradation is not taken into account [28].

In this work, to address these issues, a capacity degradation model with chemical/mechanical degradation mechanisms [17] is modified and integrated with an advanced SP model. This capacity degradation model is able to predict battery capacity loss as a function of cycle number and temperature, including SEI layer formation and growth, coupled with mechanical fatigue analysis. Further, the proposed model is able to predict voltage responses based on a physical analysis, which is an advanced method for predicting the voltage profiles as a function of cycle, compared to the existing method based on a look-up table [17,18]. The method utilizing a look-up table limited the model to the cases with well-defined tables. Also, the existing models did not consider the kinetic reactions associated with the SEI layer. In particular, the stress model in the existing approach was idealized as it was based on a single particle, which could lead to unrealistic results in certain cases. In general, the SOH of LIBs is related to capacity fade due to lithium loss and power fade due to impedance increase. Then, the capacity loss and impedance rise affect the voltage profile, which is critical information for battery management. However, most SOH estimation studies only consider the capacity change without any prediction of the voltage profile change [27–30]. Our proposed model can be directly applied to SOH estimation for battery management systems. In addition, the model is well validated with different loading conditions, including static and dynamic loadings. Further, the coupling between different physical behaviors, such as mechanical response and electrochemical reaction, is important physics in several energy related materials. The proposed approach can be applied to many other similar systems. In this paper, therefore, three unsolved challenges in the literature are addressed:

- (1) Capacity fade due to multiple degradation mechanisms and corresponding voltage change is predicted based on the physical analysis of an SP model, which does not require any lookup tables, unlike existing models.
- (2) Realistic boundaries between multiple particles are considered to calculate particle stress, which is then linked to capacity degradation. Note: the idealized single particle used for the electrochemical model is different from the single-particle based stress model in [17].
- (3) A systematic study is conducted by analyzing the relationship between the battery system's parameters (e.g., particle size, C-rate, temperature) and capacity loss due to SEI layer formation and growth on the particle and cracked surfaces.

The remainder of this paper is organized as follows. Section 2 describes the proposed methods to integrate lithium ion loss with an advanced SP model, the stress evaluation based on multiple surrounding particles, and the prediction of capacity fade due to chemical and mechanical degradation. In Section 3, the degradation model validation and the analysis of model parameter effects on capacity and power losses are given, followed by a summary and conclusions in Section 4.

2. Capacity fade mechanism modeling

2.1. Single particle model coupled with capacity degradation

Fig. 1a is a schematic overview of the simulation process in this paper. The model requires the initial battery voltage, temperature, and current input. Those initial values can be obtained from measurements. Next, the model predicts capacity fade by calculating lithium loss due to SEI layer formation and growth on the surface of particles and cracks. Then, the capacity changes are coupled with the SP model to predict the voltage profile based on the change in the lithium ion concentration. These steps are repeated for each cycle.

Fig. 1b is a schematic diagram of a lithium ion cell composed of two electrodes (a solid matrix in an electrolyte solution) and a separator (electrolyte solution), while Fig. 1c shows the concept of the SP model, which represents each electrode with a single particle. Pseudo-2D (P2D) models, which are based on porous electrode theory using concentrated solutions, are widely used for battery modeling [24,25]. This model describes lithium ion transport through the use of one-dimensional (battery thickness, x , direction) charge and mass conservation laws along with the thickness direction, as well as the diffusion process for individual active particles, which is implemented in the radial direction of the particles. The kinetic reaction at the particle surfaces is described by the Butler-Volmer equation. The P2D model consists of ten coupled nonlinear partial differential equations for the mass and charge balance in the solid and electrolyte phases [24,25].

In the SP model it is assumed that the current passing through the electrode is uniformly distributed over all of the particles inside the electrode. Therefore, as shown in Fig. 1c, each electrode can be modeled as a single spherical particle, and the governing partial differential equations can be simplified into a set of algebraic equations. Further, the complex multi-physics of the battery can be implemented in a relatively simple manner based on the assumption of uniformity of intercalation flux in the SP model [32], which leads to each reaction inside electrode, such as diffusion, transportation, diffusion-induced stress, and side-reactions, has a uniform impact on each particle. However, the existing SP models do not consider the concentration and potential distribution in electrolyte phase; therefore, model accuracy suffers at high C-rates. In order to overcome this limitation and increase model accuracy, this study uses the authors' recently developed SP model [33], in which the electrolyte physics was added.

The remaining challenge is to consider battery degradation, in particular, the coupled degradation mechanisms between mechanical crack propagation and SEI layer formation and growth. Fig. 1d and e shows the proposed SP model considering the coupling between them. In the mechanical part (Fig. 1e), the repeated stress due to the (de) intercalation is calculated, and corresponding crack propagation is predicted. In the SEI layer formation and growth model, the updated crack propagation is used to estimate the new SEI layer formation on the surface of newly-formed cracks. More details are discussed in Section 2.2.

To calculate the terminal voltage, the SP model considers the solid phase potential difference between the end sides of both electrodes based on Butler-Volmer equation [33]

$$V_t(t) = \Phi_{1,p}(t)|_{x=L} - \Phi_{1,n}(t)|_{x=0} = \left(\eta_p + \Phi_{2,p}(t)|_{x=L} + U_p \left[\frac{c_{s,p,surf}(t)}{c_{s,p,max}} \right] \right) - \left(\eta_n + \Phi_{2,n}(t)|_{x=0} + U_n \left[\frac{c_{s,n,surf}(t)}{c_{s,n,max}} \right] \right) \quad (1)$$

$$\eta_j(t) = \frac{2RT}{F} \ln[m_j(t) + \sqrt{m_j^2(t) + 1}] \quad (2)$$

$$\text{where } m_j(t) = \frac{J^{Li}}{2k_j c_{s,j,max} e^{0.5} \left[1 - \frac{c_{s,j,surf}(t)}{c_{s,j,max}} \right]^{0.5} \left(\frac{c_{s,j,surf}(t)}{c_{s,j,max}} \right)^{0.5}}$$

$$\Phi_{2,p}(t)|_{x=L} - \Phi_{2,n}(t)|_{x=0} = (1-t_+) \frac{2RT}{F} \ln \frac{c_{e,p}(L,t)}{c_{e,n}(0,t)} - \frac{I_{app}}{2} \left(\frac{L_n}{k_n^{eff}} + \frac{2L_s}{k_s^{eff}} + \frac{L_p}{k_p^{eff}} \right) \quad (3)$$

where $c_{s,j,max}$ is the maximum solid phase concentration, k_j is the reaction rate constant and $j = p/n$ denotes the positive/negative electrode, respectively, R is the universal gas constant, T is temperature, F is Faraday's constant, $\Phi_{1,j}$ is the solid-phase potential, and U_j is the Open-Circuit Potential (OCP).

The OCP, in general, is a function of the normalized surface concentration, $c_{s,j,surf}(t)/c_{s,j,max}$. The variable $k_i^{eff} = k_i \epsilon_i^{brug}$, $brug$ denotes the

Bruggeman coefficient, where k_i is the electrolyte conductivities [34]. The variable L_i is the electrode thickness, $i = p/s/n$ denotes the positive/separators/negative electrode, respectively, I_{app} is the applied current density, t_+ is cationic transport number in the electrolyte, J_j^{Li} is the electrochemical reaction rate, and $c_{e,i}(x,t)$ is the electrolyte concentration [33].

Eq. (1) does not include battery capacity reduction effects and, therefore, does not exhibit changes in the voltage profile as the battery is cycled. In order to integrate capacity loss mechanisms into the SP model, the initial concentrations, $c_{s,p}(0)|_{N+1}$ and $c_{s,n}(0)|_{N+1}$ for the cathode and anode, respectively, at the beginning of the $(N + 1)$ th cycle must be calculated. Here, one cycle is when the battery is fully discharged and then fully charged. The number of available lithium ions inside the battery after the N th cycle can be calculated based on the capacity after the N th cycle, Q_N , and the initial total number of lithium ions inside the battery, n_{Li} . From charge conservation in the anode and cathode at the $(N + 1)$ th cycle

$$\varepsilon_{s,p} L_p c_{s,p}(0)|_{N+1} + \varepsilon_{s,n} L_n c_{s,n}(0)|_{N+1} = \frac{Q_N}{Q_0} n_{Li} \quad (4)$$

where Q_0 is the battery capacity after the formation cycle (described below).

In general, battery vendors suggest a maximum voltage for charging; therefore, a scenario of full charging to that voltage was assumed in this paper. The initial voltage at the N th cycle, $V(0)$, which is constant at the beginning of each discharge cycle, is

$$U_p \left[\frac{c_{s,p}(0)|_{N+1}}{c_{s,p,max}} \right] - U_n \left[\frac{c_{s,n}(0)|_{N+1}}{c_{s,n,max}} \right] = V(0) \quad (5)$$

The initial concentrations in the anode and cathode are estimated by solving Eqs. (4) and (5). It should be noted that as the battery capacity decreases with each cycle, the corresponding OCP curve windows change, affecting the concentration polarization. The reduction in capacity can be attributed to lithium ion loss during the formation and growth of SEI layers on the surfaces of anode active material particles and cracks.

2.2. Single particle based chemical/mechanical degradation model

2.2.1. Capacity degradation

A schematic diagram of SEI formation, growth, and additional formation and its growth due to crack propagation is shown in Fig. 1d and e. A fresh battery cell (Stage I) undergoes a process called ‘formation cycle’ to stabilize the cell by forming an initial SEI layer. During the formation cycle, a uniform SEI layer, mainly containing Li_2O , LiF , and Li_2CO_3 , is formed on the freshly exposed particle surfaces and initial crack surface due to electrolyte reduction (Stage II). Once the battery is in use, the SEI layer tends to grow, especially when the battery is used at high temperatures (Stage III). In addition, the diffusion-induced stress due to changes in the battery volume causes cracks to propagate on the particle surface, creating new surfaces that are exposed to the electrolyte and on which new SEI layers will form. This repeated process continuously consumes lithium ions, causing capacity fade as shown Fig. 1f. The growth model developed by Rutooj et al. [17] is utilized in this work and coupled with the SP model given in this paper to include degradation mechanisms due to SEI layer formation and growth.

Solid electrolyte interface layer formation and growth can be categorized into four different mechanisms including: (1) initial SEI layer formation during the formation cycle; (2) growth on the initially formed SEI layer; (3) initial SEI layer formation on newly formed surfaces due to cracking; and (4) growth of the initially formed SEI layers on the surfaces created by cracking.

During the formation cycle, the particle will be fully covered with an SEI layer. The initial SEI layer thickness, L_{SEI}^0 , can be estimated based

on the capacity loss after the formation cycle, which is assumed to be 10% [5,17]

$$L_{SEI}^0 = \frac{0.1 Q_{in} M_{SEI}}{A_{initial} n_{SEI} \rho_{SEI} F} \quad (6)$$

where the Q_{in} is the battery capacity before the SEI layer formation cycle process is conducted, $A_{initial}$ is the particle surface area before the SEI layer formation cycle, n_{SEI} is the number of lithium moles lost for every mole of SEI layer formed, ρ_{SEI} is the SEI layer density, and M_{SEI} is the molecular weight of compounds constituting the SEI layer.

To evaluate the surface area of the SEI layer, defects must be considered that exist in active material particles and will be the seeds that initiate cracks. The initial surface area of the single particle is the surface area of a sphere with initial defects [17]

$$A_{initial} = 4\pi r_n^2 (1 + 2\rho_{cr} l_{cr0} a_0) \quad (7)$$

where l_{cr0} is the initial defect width, a_0 is the initial defect length, r_n is the particle radius, and ρ_{cr} is the number of cracks per unit area.

Since SEI layer growth generally follows a parabolic pattern, SEI layer thickness, L_{SEI} , growth rate per cycle number (N) can be expressed as

$$\frac{dL_{SEI}}{dN} = \frac{1}{2} K_{SEI} N^{-\frac{1}{2}} \quad (8)$$

where $K_{SEI} = K_{SEI0} e^{-\frac{E_{a2}}{RT}}$, E_{a2} is the activation energy for SEI layer thickness growth, and K_{SEI0} is the SEI layer growth coefficient [17].

The SEI layer will grow on the initial particle surface, thus, combining Eqs. (6)–(8), the volume change of the SEI layer growth on the initial SEI layer, i.e., the second degradation mechanism, is

$$\frac{dV_{SEI,2}}{dN} = 2\pi r_n^2 (1 + 2\rho_{cr} l_{cr0} a_0) K_{SEI} N^{-\frac{1}{2}} \quad (9)$$

As the number of battery cycles increases, not only does the formed SEI layer continue to grow, the initial defects also grow and form cracks on the surfaces of the particles due to repeated stress on the active material particles. These cracks result in new interface area with the electrolyte. Paris’ law is used here to model such crack propagation [35]

$$\frac{da}{dN} = k (\sigma_{\theta,cor} b \sqrt{\pi a})^m \quad (10)$$

where a is the crack length and b are empirical constants that depend on material properties [36]. The crack propagation coefficient can be expressed based on the Arrhenius form, $k = k_0 e^{-\frac{E_{a1}}{RT}}$, where E_{a1} is the activation energy for crack propagation and k_0 is the crack propagation coefficient [17].

In order to determine the crack growth rate, the maximum cyclic stress on the particle surface is used. Most studies concerning stress analysis in battery models are based on a single isolated particle under a steady-state condition and calculate the maximum tensile stress to be [37]

$$\sigma_{\theta,max} = \frac{-E \Omega I_{app}}{45(1-\nu) \varepsilon_n l_n} \left(\frac{r_n^2}{F D_{s,n}} \right)^2 \quad (11)$$

where E is Young’s modulus, ν is Poisson’s ratio of the electrode material, and Ω is the partial molar volume of the solute. This model computes the stress levels in a single particle caused by diffusion; however, it does not consider the constraint imposed on a particle’s boundary when it is in contact with other particles. This analysis is extended in this paper to a more realistic situation where there is a network of particles and calculates the maximum stress at the particle surface based on the strain during charge and discharge.

The driving force for lithium ion diffusion can be obtained by the gradient of the concentration and mechanical strain energy and, correspondingly, the mass conservation of lithium ions becomes [11]

$$\frac{\partial c_{sj}(r,t)}{\partial t} = D_{sj} D \left[\nabla^2 c_{sj}(r,t) - \frac{\Omega}{RT} \nabla c_{sj}(r,t) \cdot \nabla \sigma_h - \frac{\Omega c_{sj}(r,t)}{RT} \nabla^2 \sigma_h \right] \quad (12)$$

where σ_h is the hydrostatic stress of a spherical particle [11]. The stress-strain relationship considering the effect of intercalation is [38,39]

$$\varepsilon_{mn} = \frac{1}{E} [(1 + \nu) \sigma_{mn} - \nu \sigma_{kk} \delta_{mn}] + \frac{(c_{sj}(r,t) - c_{sj}(r,0)) \Omega}{3} \delta_{mn} \quad (13)$$

where ε_{mn} is the components of strain tensor ($m, n, k = 1, 2, 3$), σ_{mn}/σ_{kk} are components of stress tensor, and δ_{mn} is the Kronecker delta.

The stress acting on a particle is now calculated based on a network of particles, and the particle surface stress ratios are normalized based on the stress of a single particle with a radius of 5 μm [17]. The schematic of a network of 5 particles is shown in Fig. 2a and the stress increase ratio for networks with 3–7 particles is shown in Fig. 2b as a function of particle size. As shown in Fig. 2b, when using a network of particles the stress increases linearly with particle size, which is in stark contrast to the quadratic relationship between the stress and particle radius for a single particle as given in Eq. (11). Comparing to a network of seven particles, the stress ratio differences for 3–6 particles are 9.58%, 3.77%, 1.45%, and 1.3%, respectively; therefore, a network of 5 particles is used in this study. The stress equation for a single particle (Eq. (11)) is calibrated using a correction factor, β , which is based on the stress calculation with a network of 5 particles divided by the stress calculated from Eq. (11). The corrected maximum stress is

$$\sigma_{\theta,cor} = \beta \left[\frac{-E \Omega l_{app}}{45(1-\nu) \varepsilon_n l_n} \left(\frac{r_n^2}{F D_{s,n}} \right)^2 \right] \quad (14)$$

The total area change associated with crack propagation due to stress for each cycle is

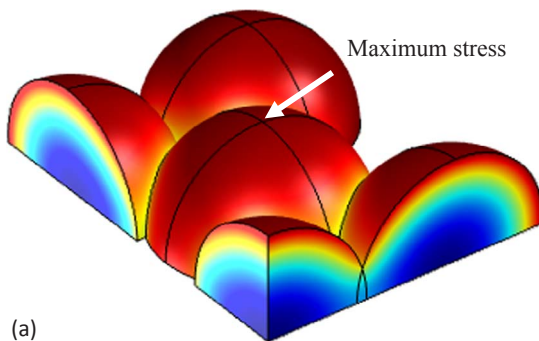
$$\frac{dA_{cr}}{dN} = 8\pi r_n^2 \rho_{cr} l_{cr0} k (\sigma_{\theta,cor} b \sqrt{\pi a_0})^m \left(\frac{a}{a_0} \right)^{\frac{m}{2}} \quad (15)$$

Correspondingly, the volume change of the SEI layer formation at newly exposed surfaces per cycle, i.e., the third degradation mechanism, is

$$\frac{dV_{SEI,3}}{dN} = L_{SEI}^0 8\pi r_n^2 \rho_{cr} l_{cr0} k (\sigma_{\theta,cor} b \sqrt{\pi a_0})^m \left(1 + \frac{2-m}{2} k [\sigma_{\theta,cor} b \sqrt{\pi}]^m a_0 \left(\frac{m}{2-m} \right) N \right)^{\frac{m}{2-m}} \quad (16)$$

The growth rate of the SEI layer formed on cracks as they grow, i.e., the fourth degradation mechanism, is

$$\frac{dV_{SEI,4}}{dN} = \sum_{i=1}^{N-1} \left(8\pi r_n^2 \rho_{cr} l_{cr0} k (\sigma_{\theta,cor} b \sqrt{\pi a_0})^m \left(1 + \frac{2-m}{2} k [\sigma_{\theta,cor} b \sqrt{\pi}]^m a_0 \left(\frac{m}{2-m} \right) N \right)^{\frac{m}{2-m}} \right)_i \left(\frac{1}{2} K_{SEI} N^{-\frac{1}{2}} \right)_{N-i} \quad (17)$$



Finally, the total SEI layer growth rate is

$$\frac{dV_{SEI}}{dN} = \frac{dV_{SEI,2}}{dN} + \frac{dV_{SEI,3}}{dN} + \frac{dV_{SEI,4}}{dN} \quad (18)$$

Capacity decay can be calculated based on SEI layer growth. The initial battery capacity of a fresh cell, Q_{in} in Eq. (6), is estimated to be

$$Q_{in} = q_r Q_g \left(\frac{4}{3} \pi r_n^2 \rho_g \right) \quad (19)$$

where Q_g is the specific capacity of graphite, ρ_g is the density of graphite, and q_r is a formation cycle efficiency, which is assumed to be 0.9 [17]. The battery capacity after the SEI formation cycle is

$$Q_0 = 0.9 Q_{in} \quad (20)$$

Then, the capacity fade rate per cycle after the formation cycle is

$$\frac{dQ_N}{dN} = - \frac{n_{SEI} \rho_{SEI} F}{M_{SEI}} \frac{dV_{SEI}}{dN} \quad (21)$$

2.2.2. Resistance due to SEI layer formation and growth

The SEI layer growth may also increase the cell resistance and impact battery power. Battery resistance increase is estimated by leveraging the work of Ning et al. [19]. The SEI layer resistance at the N th cycle is

$$R_{SEI} \ln = \frac{L_{SEI} \ln}{k_{SEI}} \quad (22)$$

where k_{SEI} is the SEI layer conductivity and the SEI layer thickness at the N th cycle is

$$L_{SEI} \ln = \frac{V_{SEI} \ln}{A_{cr} \ln} \quad (23)$$

The initial SEI layer resistance, R_{SEI}^0 , is assumed to be on the same order as the electrolyte resistance [40]. Thus, k_{SEI} is calculated based on the initial SEI layer thickness and resistance

$$k_{SEI} = \frac{L_{SEI}^0}{R_{SEI}^0} \quad (24)$$

Finally, the cell terminal voltage, including the impact of initial concentration change (Eqs. (4) and (5)), SEI layer resistance (Eq. (22)), and other polarizations effects such as activation and diffusion, is

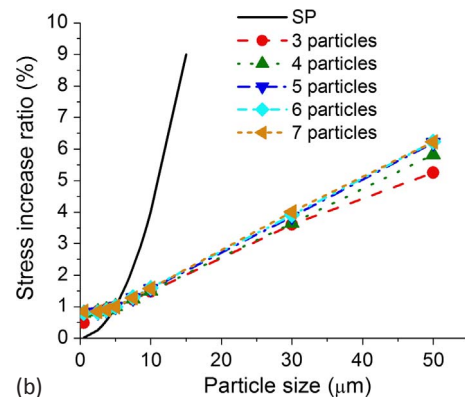


Fig. 2. (a) Schematic of five particles and (b) particle surface stress ratio as a function of particle size for various numbers of particles.

Table 1
Model parameters used in simulation studies.

Parameter	Value	Description	Ref.
a_0	2×10^{-9}	Initial crack length (m) [17]	[17]
$brug$	1.5	Bruggeman coefficient	[17]
c_0	1000	Initial concentration (mol m^{-3})	[17]
$C_{\text{max, pos}}$	22,806	Positive maximum concentration (mol m^{-3})	[17]
$C_{\text{max, neg}}$	30,555	Negative maximum concentration (mol m^{-3})	[17]
D_e	7.5×10^{-11}	Diffusion coefficient in electrolyte ($\text{m}^2 \text{s}^{-1}$)	[17]
$D_{s,n}$	10×10^{-14}	Solid-phase Li diffusivity, negative electrode ($\text{m}^2 \text{s}^{-1}$)	[17]
$D_{s,p}$	5.9×10^{-20}	Solid-phase Li diffusivity, positive electrode ($\text{m}^2 \text{s}^{-1}$)	[17]
E_{a1}	10.1	Activation energy for crack propagation (kcal mol^{-1})	Fitted
E_{a2}	5.791	Activation energy for SEI layer growth (kcal mol^{-1})	Fitted
E	3.3×10^{10}	Young's modulus of electrode material (N m^{-2})	[17]
F	96,487	Faraday's constant (C mol^{-1})	
k_j	2×10^{-11}	Reaction rate constant ($\text{m}^{2.5} \text{mol}^{-0.5} \text{s}^{-1}$)	[17]
k_0	1.62×10^{-16}	Crack propagation coefficient	Fitted
K_{SEI0}	7.195×10^{-11}	SEI layer growth coefficient	Fitted
l_{cr0}	2×10^{-9}	Initial crack width (m)	[17]
L_n	3.8×10^{-5}	Negative electrode thickness (m)	[17]
L_p	8×10^{-5}	Positive electrode thickness (m)	[17]
L_s	2.5×10^{-5}	Separator thickness (m)	[17]
M_{SEI}	78.89	Molecular weight of compounds constituting SEI (g mol^{-1})	[17]
n_{SEI}	2	Consumed lithium ion for 1 mol SEI layer formation	[17]
Q_g	0.339	Graphite specific capacity (A h g^{-1})	[17]
R	8.314	Universal gas constant ($\text{J mol}^{-1} \text{K}^{-1}$)	
r_n	5×10^{-6}	Particle radius, negative electrode (m)	[17]
r_p	5×10^{-8}	Particle radius, positive electrode (m)	[17]
R_{SEI}^0	1×10^{-3}	Initial SEI layer resistance (Ω)	Assumed [39]
t_+	0.363	Cationic transport number	
$V(0)_N$	4.2	Initial voltage at each cycle (V)	
ε_n	0.248	Negative electrode porosity	[17]
ε_p	0.367	Positive electrode porosity	[17]
ε_s	1	Separator porosity	[17]
ρ_{cr}	2.542×10^{18}	Number of cracks per unit area of particle (m^{-2})	[17]
ρ_g	2.26	Graphite density (g cm^{-3})	[17]
ρ_{SEI}	2.11×10^6	Density of SEI films (g m^{-3})	[17]
Ω	8.9×10^{-6}	Partial molar volume of solute ($\text{m}^3 \text{mol}^{-1}$)	[17]

$$\begin{aligned}
 V_i(t)_{N+1} = & U_p \left[\frac{c_{s,p,\text{surf}}(t)_{N+1}}{c_{s,p,\text{max}}} \right] - U_n \left[\frac{c_{s,n,\text{surf}}(t)_{N+1}}{c_{s,n,\text{max}}} \right] \\
 & + \frac{2RT}{F} \{ \ln[m_p(t) + \sqrt{m_p^2(t) + 1}] - \ln[m_n(t) + \sqrt{m_n^2(t) + 1}] \} \\
 & + (1-t_+) \frac{2RT}{F} \ln \frac{c_{e,p}(L,t)}{c_{e,n}(0,t)} - \frac{I_{app}}{2} \left(\frac{L_n}{k_n^{\text{eff}}} + \frac{2L_s}{k_s^{\text{eff}}} + \frac{L_p}{k_p^{\text{eff}}} \right) \\
 & + R_{\text{SEI}} \ln I_{app}.
 \end{aligned} \quad (25)$$

In order to demonstrate the effectiveness of the developed model, the LiFePO_4/C battery chemistry is adopted in this work. The parameters used in the simulation studies conducted in this paper are listed in Table 1.

3. Results and discussion

3.1. Validation of degradation model

3.1.1. Capacity degradation under a constant loading

To verify the proposed model, predicted battery capacity is compared with previously observed experimental data for a $\text{LiFePO}_4/\text{Graphite}$ battery [41] as a function of cycle number and temperature from 15 to 60 °C for a discharge rate of 0.5 C. As shown in Fig. 3a, the simulated capacity fade matches the measured data well with fractional capacity Root Mean Square Errors (RMSEs) of 7.21×10^{-3} , 7.43×10^{-3} , and 10.3×10^{-3} at 15, 45, and 60 °C, respectively. As expected, the capacity fades at a faster rate as the temperature increases. Despite its low-order structure, the developed model is in good agreement with the experimental data, indicating that the proposed capacity fade model based on single particle assumptions can describe

the degradation of lithium ion batteries at different temperatures.

Accurate voltage profile prediction as the battery goes through cycles of charging and discharging is a critical function of the BMS. The proposed model predicts the voltage profile based on the physics described in Eqs. (4), (5), and (25), which is different from the work in [17] that relied on a look-up table. Lithium ion losses due to SEI layer formation and growth affect not only the available battery capacity, but also the operating voltage window due to shifts in the OCP curves. Fig. 3b and c shows that the range of the anode (Li_xC_6) OCP curve shifts to the left due to lithium ion losses. On the other hand, the cathode (Li_xFePO_4) OCP curve does not shift despite the loss of lithium ions due to the unique long plateau of this curve. As a result of the coupling between the anode and cathode, the voltage profiles vary with the cycle number. The simulation results are in good agreement with an RMSE on the order of 1×10^{-2} (Table 2) compared to the experimental voltage profile in [41,42]. Additionally, the simulation is able to capture the same behavior as the experimental observations in which they observed that the anode OCP window shifted to the left as the battery capacity decreased, and the cathode OCP window did not shift [17,18].

Capacity fade is mainly due to the lithium ion loss resulting from the four mechanisms described above. The first mechanism (i.e., initial SEI layer formation) occurs before the battery is put into operation. Next, the effect of the other three degradation mechanisms is analyzed in details. Fig. 3d shows the contributions of mechanisms 2–4 as a ratio of the SEI volume formed due to each mechanism to the initial SEI volume formed during the formation cycle (i.e., mechanism 1). The newly-formed SEI volume ratio due to new crack surfaces (i.e., mechanism 3) reaches approximately 9% after 900 cycles, which is when the battery reaches 80% of its initial capacity, with a rate of 9.76×10^{-5} volume ratio per cycle, while the volume increase due to the growth of the

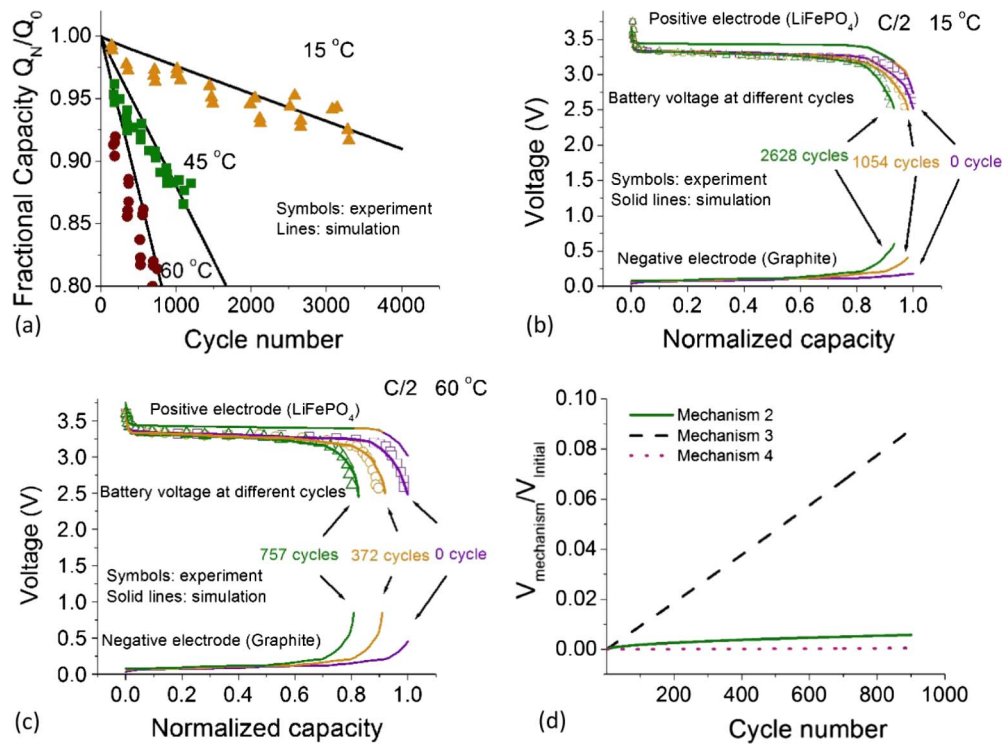


Fig. 3. (a) Cell fractional capacity as a function of number of cycles with 0.5 C, and simulation and experimental results for LiFePO₄/Graphite battery: open circuit voltages and voltage profiles for (b) 15 °C and (c) 60 °C, and (d) SEI layer volume ratio $V_{mechanism}/V_{initial}$ for mechanism 2 SEI layer growth on the initial SEI layer, mechanism 3 SEI layer formation on new crack surfaces, and mechanism 4 SEI layer growth on SEI layers formed on crack surfaces as a function of cycle number.

Table 2
Root Mean Square Errors of voltage profile at C/2 with 15 °C and 60 °C.

	0th cycle	1054th cycle	2628th cycle
C/2 and 15 °C	54.4×10^{-3}	39.9×10^{-3}	35.0×10^{-3}
	0th cycle	372th cycle	757th cycle
C/2 and 60 °C	24.5×10^{-3}	57.8×10^{-3}	44.2×10^{-3}

initial SEI layer (i.e., mechanism 2) and the SEI volume formation due to growth on newly-formed crack surfaces (i.e., mechanism 4) are less than 1% after 900 cycles with rates of 5.25×10^{-6} and 5.68×10^{-7} volume ratio per cycle, respectively. Therefore, SEI layers formed on the cracks dominate the battery degradation.

3.1.2. Capacity fade under dynamic loading conditions

The SEI layer formation mainly occurs during the charging process and, for typical EVs, batteries are charged using a Constant Current/

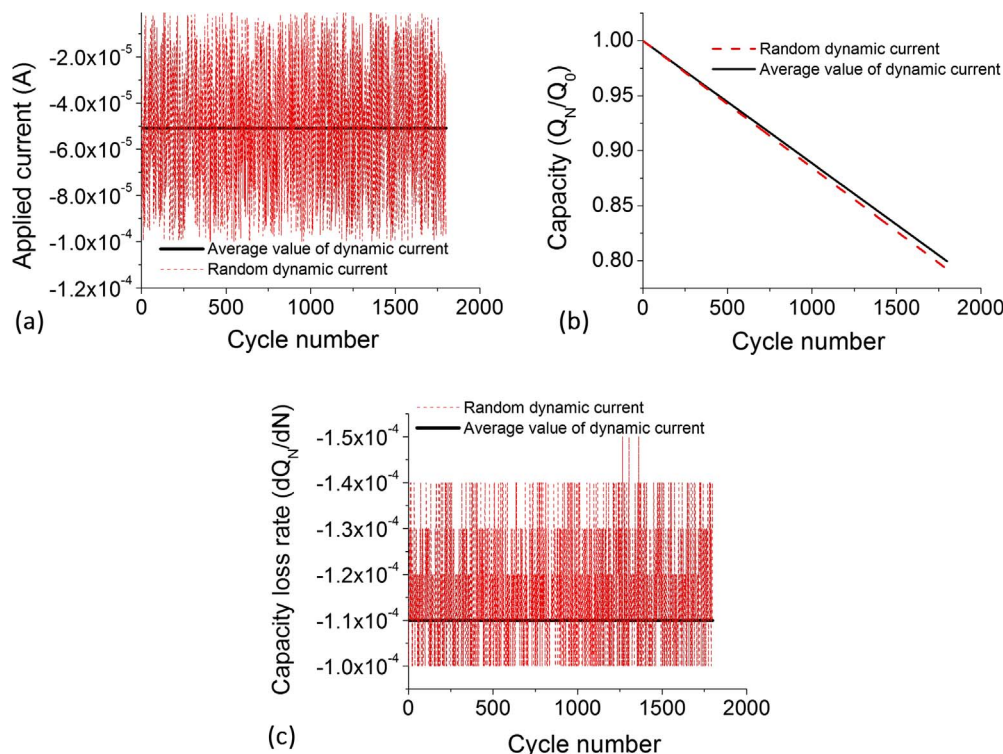


Fig. 4. Battery degradation results: (a) dynamic and constant average applied currents, (b) cycle history of normalized capacity prediction for dynamic and constant applied currents, and (c) cycle history of capacity change cycle for dynamic and constant applied currents.

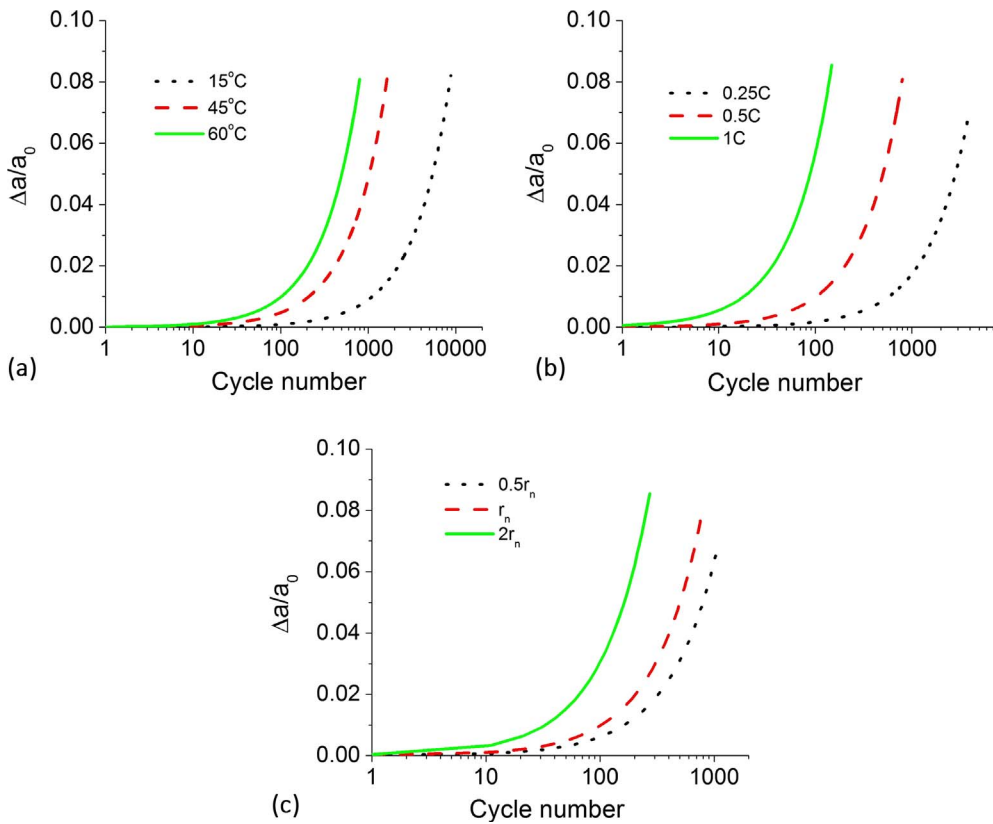


Fig. 5. Crack propagation $\Delta a/a_0$ versus cycle number (a) different temperatures with 0.5 C and r_n , (b) different C-rates with 60°C and r_n , and (c) different particle sizes with 0.5 C and 60°C.

Constant Voltage (CCCV) mode. Section 3.1.1 focused on a constant current case, while this section considers a case in which the charging current may be different at each cycle. For this dynamic loading condition, a random current loading (from 0 to 0.5 C-rate) is applied during charging at each cycle at 25°C. Battery degradation results are shown in Fig. 4 for a dynamic current profile and the corresponding average constant current profile. As shown in Fig. 4b, the capacity fade when subjected to dynamic loading results in greater capacity fade (0.7% greater at 80% capacity loss) than when subjected to the dynamic profile's average current. This phenomenon, in which a dynamic current produces a faster degradation rate (1.16×10^{-4} per cycle) than the constant current (1.10×10^{-4} per cycle) as shown in Fig. 4c, was observed in experimental data [43] for an electric vehicle battery under highway driving loading profiles. In this experiment, the capacity fade with highway driving load profiles is slightly greater (approximately 0.1%) than the case with a constant current.

3.2. Capacity loss and power loss analyses

Crack propagation, $\Delta a/a_0$, defined as the ratio of the change in crack length to the initial crack length, is plotted in Fig. 5 as a function of temperature, C-rate, and particle size (with $\beta = 3.3$, 1, and 0.4 for $0.5r_n$, r_n , and $2r_n$, respectively). As shown in this figure, all of the parameters have a significant impact on crack propagation and, hence, battery cycle life. Lower temperatures can slow the growth of the SEI layer (Eq. (8)), and lower C-rates or smaller particle sizes can lower the stress level (Eq. (14)), both of which slow the crack growth rate (Eq. (10)) and, correspondingly, extend battery life. There is a lot of experimental evidence that stress can build up inside battery materials and there are also some experimental studies characterize the developed stress and cracks based on measurements from SEM and AFM [41,44–47]. However, in all of the prior work, the crack growth inside particles was not monitored and, practically, it will be very hard to monitor stress at the particle level based on in-situ measurement. The predicted crack propagation rate in this work are in the range of values from the reference [41,44–46].

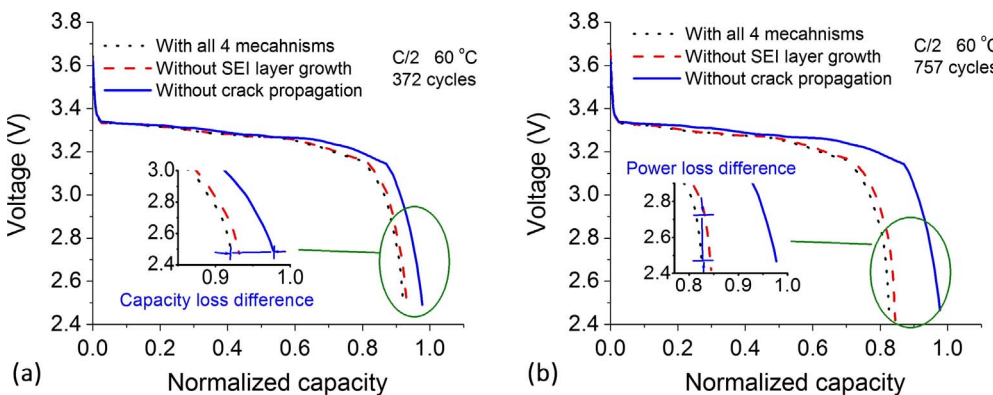


Fig. 6. Voltage profiles and resulting capacity fade and power loss when considering a variety of degradation mechanisms at (a) 372 and (b) 757 cycles.

Fig. 6 shows examples of voltage profiles for three cases at two specific cycle numbers, $N = 372$ (Fig. 6a) and $N = 757$ (Fig. 6b). Case 1 does not consider crack propagation, i.e., degradation mechanisms 3 and 4 (solid line) are not included. Case 2 does not consider SEI layer growth on previously formed SEI layers, i.e., degradation mechanisms 2 and 4 (dashed line) are not included. Case 3 considers all four degradation mechanisms (dotted line). As shown in Fig. 6, capacity reduction and power loss are slightly less for Case 2 as compared to Case 1. However, in the absence of crack propagation (Case 1), there are significant differences in capacity fade and power loss as compared to Case 3. Details of capacity fade and power loss differences are discussed below.

3.2.1. Capacity loss analysis

Capacity Fade Deviation (CFD) percentage is defined as

$$CFD = \frac{Q_{case,3} - Q_{case,i}}{Q_{case,3}} \times 100 \quad (26)$$

where $Q_{case,3}$ is the calculated capacity fade for Case 3 and $Q_{case,i}$ is the calculated capacity fade for Case 1 or Case 2. Therefore, the greater the CFD, the greater the influence of crack propagation (Case 1) or SEI layer growth (Case 2) on battery capacity fade. Compared to the results that

take all degradation mechanisms into account, the results for Case 2 show little change in the capacity fade, the CFD values are 0.9% and 1.8% at 372 and 757 cycles, respectively. In contrast, there is a relatively large difference in the capacity fade, i.e., the CFD values are 6.2% and 18.3% at 372 and 757 cycles, respectively, for Case 1. This result indicates that crack propagation accelerates SEI layer formation and, thus, capacity fade, significantly.

Issues such as crack propagation, SEI layer growth, capacity fade, and power loss can be affected by other factors, such as C-rate, temperature, and particle size. Additional studies were conducted to explore these effects and the results are shown in Fig. 7, which includes the impact of temperature (Fig. 7a and b), C-rate (Fig. 7c and d), and particle size (Fig. 7e and f). The results in Fig. 7a, c, and e exclude the effect of crack propagation and the results in Fig. 7b, d, and f exclude the effect of SEI layer growth. All of the simulations are terminated assuming battery life is over when the battery capacity reaches 80% of the initial capacity.

Temperature is an important factor determining battery life, as it directly affects the SEI layer formation rate (Eq. (8)) and the crack propagation rate (Eq. (10)). As shown in Fig. 7a and b, for the original particle size ($5\mu\text{m}$) at 0.5 C, higher temperatures increase the crack growth rate and SEI layer growth rate, resulting in more SEI layer

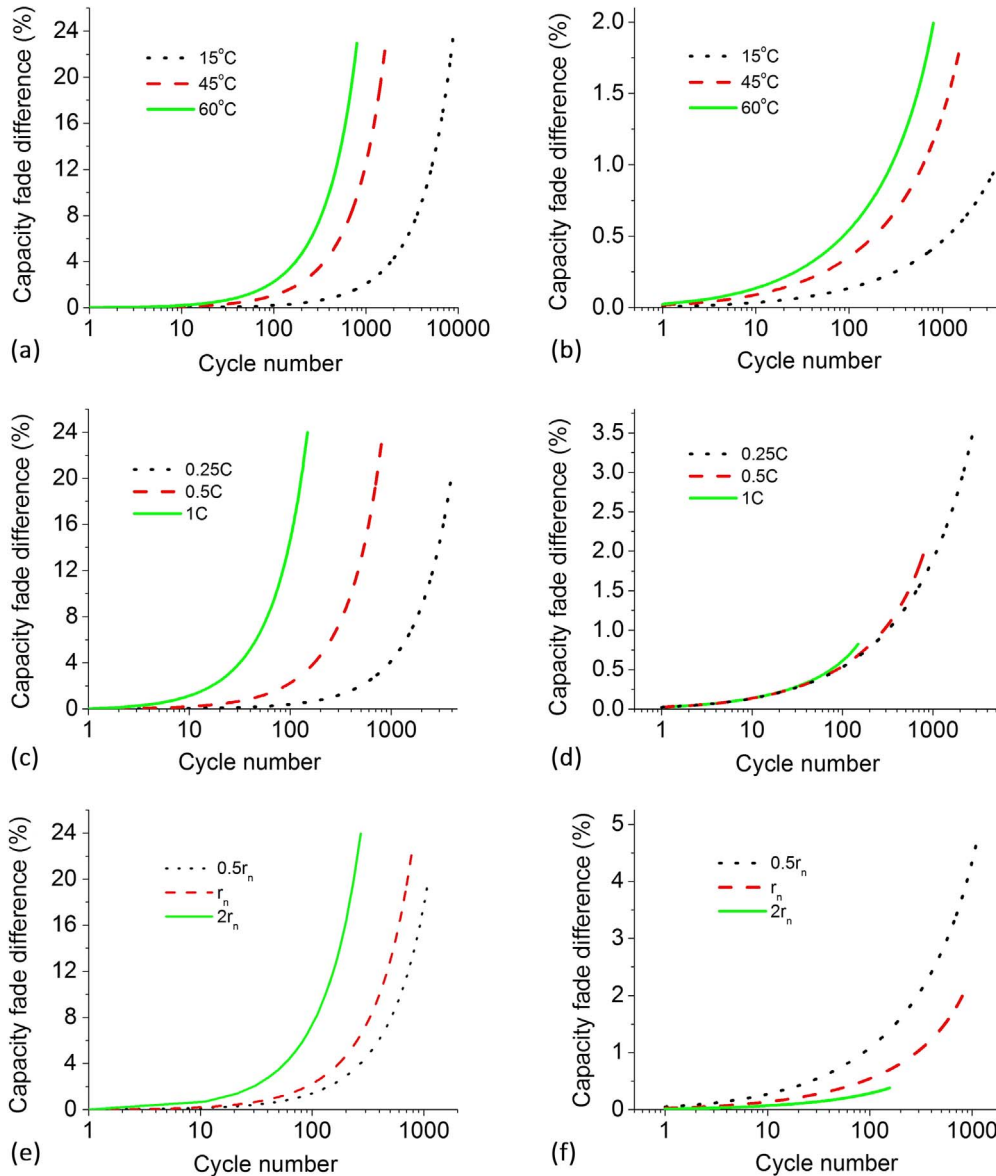


Fig. 7. Capacity fade difference between Cases 1 and 3 at (a) different temperatures with 0.5 C and r_n , (c) different C-rates with 60 °C and r_n , and (e) different particle sizes with 0.5 C and 60 °C. Capacity fade difference between Cases 2 and 3 at (b) different temperatures with 0.5 C and r_n , (d) different C-rates with 60 °C and r_n , and (f) different particle sizes with 0.5 C and 60 °C.

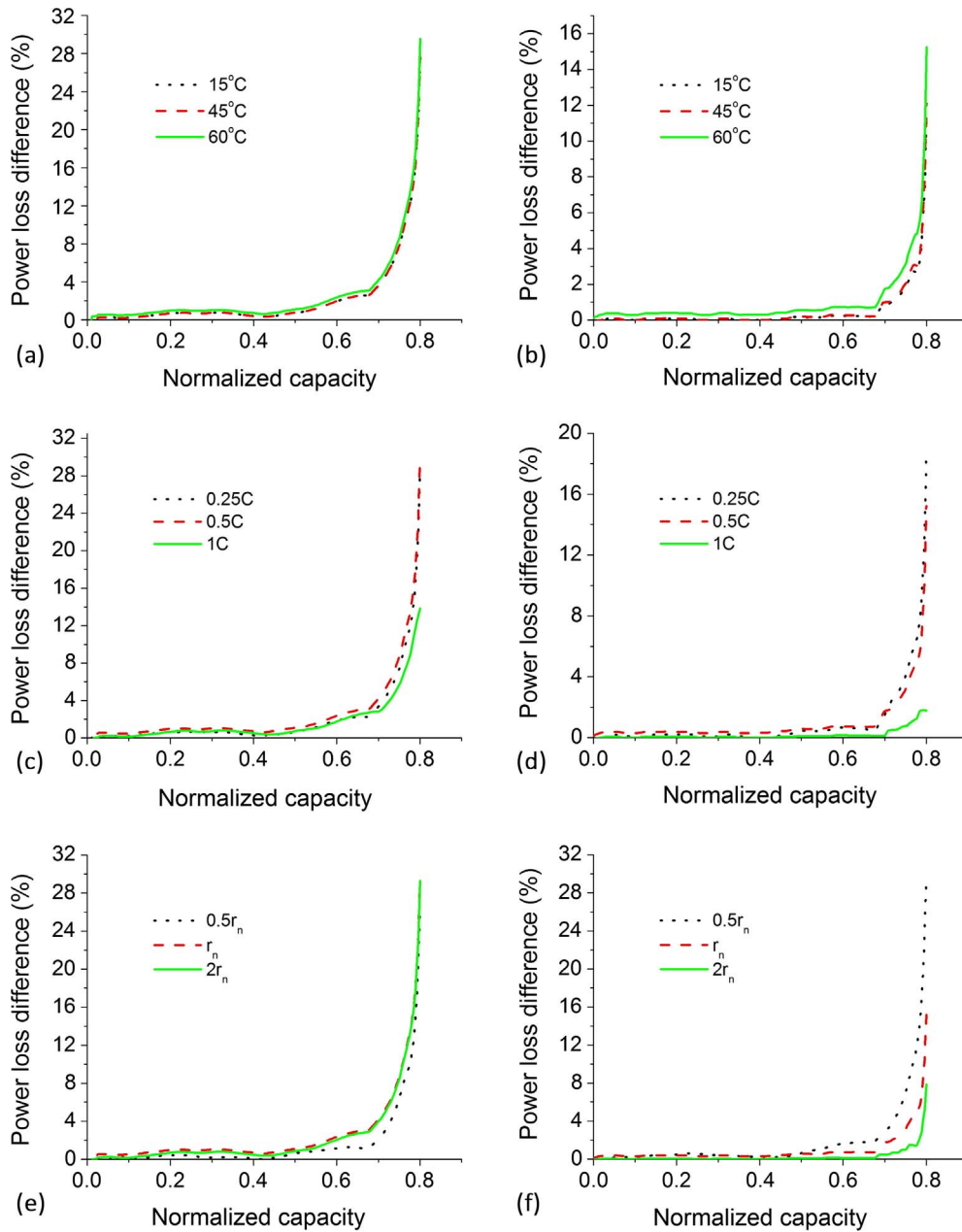


Fig. 8. Power loss differences between Cases 1 and 3 at (a) different temperatures with 0.5 C and r_n , (c) different C-rates with 60 °C and r_n , and (e) different particle sizes with 0.5 C and 60 °C. Power loss differences between Cases 2 and 3 at (b) different temperatures with 0.5 C and r_n , (d) different C-rates with 60 °C and r_n , and (f) different particle sizes with 0.5 C and 60 °C.

Table 3
Maximum CFD and PLD values for different conditions.

Parameters	Capacity fade deviation		Power loss deviation	
	No crack propagation (%)	No SEI layer growth (%)	No crack propagation (%)	No SEI layer growth (%)
15 °C, 0.5 C, r_n	23.29	1.01	27.42	11.36
45 °C, 0.5 C, r_n	23.06	1.78	27.64	12.12
60 °C, 0.5 C, r_n	22.95	1.99	29.53	15.25
60 °C, 0.25 C, r_n	20.32	3.53	28.18	18.42
60 °C, 1 C, r_n	23.99	0.82	13.86	1.77
60 °C, 0.5 C, $0.5r_n$	19.80	4.77	25.90	19.52
60 °C, 0.5 C, $2r_n$	23.98	0.38	29.29	7.86

formation on the cracked surfaces. Case 1 shows the CFD values are 23.29% and 22.95% at 15 and 60 °C, respectively (Fig. 7a); however, the results of Case 2 show that the CFD values are 1.01% and 1.99% at

15 and 60 °C, respectively (Fig. 7b).

The C-rate impacts capacity fade because it affects not only the battery electrochemical reaction (Eq. (1)) but also the maximum stress on the particle surface (Eq. (14)). As shown in Fig. 7c and d, a higher C-rate results in shorter battery life. For instance, the cycle life is 4000 cycles at 0.25 C while the life is only 150 cycles at 1 C. This is due to the fact that higher C-rates promote crack propagation by increasing the maximum stress at the particle surface. In Case 1, 1 C results in a CFD value of 23.99%, which is 4.19% greater than the case for 0.25 C. Additionally, the results from Case 2 for all of the C-rates show smaller values of CFD than the corresponding results for Case 1.

Particle size will also impact capacity fade because it affects the maximum stress at the particle surface (Eq. (14)) and the total surface area (Eq. (7)). Larger particles undergo greater volume change, causing higher stresses. As a result, the CFD value is much greater for Case 1, as compared to the CFD values for Case 2 at all C-rates. As shown in Fig. 7e and f, the deviation of larger particles ($2r_n$, 23.98% CFD) used in Case 1 is 4.18% greater for the smaller particle ($0.5r_n$, 19.80% CFD); however, the larger particles ($2r_n$, 0.38% CFD) used in Case 2 have a

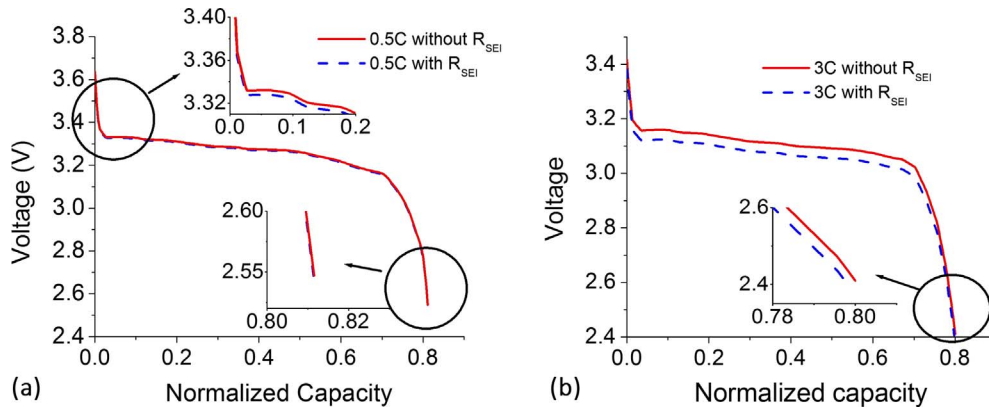


Fig. 9. Voltage profiles with and without consideration of SEI layer resistance at 60 °C for (a) 0.5 C and (b) 3 C after 757 cycles.

value of CFD that is 4.33% smaller than the CFD value for the smaller particle ($0.5r_n$, 4.77% CFD), which could be explained by higher current density on the smaller particles.

3.2.2. Power loss analysis

Power is another important measure of battery performance. Similar to capacity fade deviation, a Power Loss Deviation (PLD) percentage is defined as

$$PLD = \frac{I_{app} V_{case,i} - I_{app} V_{case,3}}{I_{app} V_{case,3}} \times 100 = \frac{V_{case,i} - V_{case,3}}{V_{case,3}} \times 100 \quad (27)$$

where $V_{case,3}$ is the calculated voltage for Case 3 and $V_{case,i}$ is the calculated voltage for Cases 1 or 2.

The PLD values were calculated based on the voltage profiles changes (Fig. 6) at different operating conditions and the results are given in Fig. 8 for a variety of temperatures, C-rates, and particle sizes. As can be seen in Fig. 8, the PLDs indicate that the voltages for Cases 1 and 2 are always greater than the voltage for Case 3. In addition, the PLD value is always significantly larger for higher normalized capacities. This means that the voltage profile changes significantly due to the capacity fade at the end of discharge, which is related to a high concentration polarization resulting from the shift of the OCP window as mentioned above.

The effects of temperature on the voltage loss were similar to the effects of temperature on the capacity fade. Similar to the CFD results for Case 2 where low C-rates caused increased capacity fade due to prolonged battery cycling life, low C-rates will also cause increased battery power loss. In addition, the PLD values for Case 1 (Fig. 8c) have 28.18% and 29.53% differences at the end of discharge for 0.25 and 0.5 C, respectively, similar to the trend of CFD with C-rates. The PLD value for Case 1 at 1 C (13.86%), however, is much smaller than the CFD value for Case 1 (23.99%), which could be due to the fact that the total number of lithium ion loss is not sufficient to affect the voltage profile as calculated by Eqs. (4) and (5). For particle sizes of $0.5r_n$, r_n , and $2r_n$, the results also generally follow the CFD trends for both Cases 1 and 2.

In summary, the maximum CFD and PLD values at different conditions in Table 3 demonstrate that the effect of SEI layer growth on capacity fade strongly depends on temperature (Fig. 7a and b), C-rate (Fig. 7c and d), and particles size (Fig. 7e and f). Also, the effect of SEI layer growth on power loss strongly depends on C-rate (Fig. 8c and d) and particle size (Fig. 8e and f), but it is less sensitive to temperature (Fig. 8a and b). The effect of crack propagation on capacity fade decreases as the C-rate and particle size decrease. It also can be concluded that the effect of crack propagation is more important than the effect of SEI layer growth on capacity fade and power loss for all of the operating conditions considered here.

3.2.3. SEI layer resistance

The SEI layer can interfere with ion transport and may result in increased internal battery resistance. In this study, the SEI layer resistance is integrated with the SP model via Eqs. (22)–(25) to study the effect of SEI layer resistance on the voltage curve profile and capacity fade. Fig. 9 shows the voltage profiles for two different C-rates with and without considering SEI layer resistance, R_{SEI} , in Eq. (25). At 0.5 C, the effect of SEI layer resistance on voltage drop at the start of discharge is very small (0.1%) and has no effect on the final capacity, which is consistent with experimental observations at 0.5 C [40]. However, as shown in Fig. 9b, the initial voltage drop increases by 2% (0.03 V) and shows a 0.3% capacity fade at 3 C when the capacity decayed to 80%. Therefore, it can be concluded that SEI layer resistance has a greater impact on the voltage profile as the C-rate increases.

4. Summary and conclusions

In this paper, an advanced capacity fade model, coupled with chemical and mechanical degradation mechanisms, was developed based on an SP model. This SP model includes electrolyte physics, alleviating the problem of low accuracy at high C-rates. The proposed model was able to accurately predict battery capacity fade and voltage profile as a function of cycle number over a broad temperature range with an error of 10.3×10^{-3} RMSE compared to experimental results. The key findings can be summarized as follows:

- (1) The SEI layer formation on newly formed surfaces due to the crack propagation dominates the capacity degradation during cycling; it had a 9 times greater effect on capacity fade than the growth of the existing SEI layer on all surfaces (i.e., particle and cracks).
- (2) The effect of crack propagation on capacity strongly depends on temperature, C-rate, and particle size, and the effect of crack propagation on power loss increases at higher temperatures, lower C-rates, and smaller particle sizes.
- (3) Battery resistance increased slightly (0.1%) with cycling at low C-rate (0.5 C) and had minimal impact on battery capacity fade, which agrees well with the experimental observations from the literature. At higher C-rates (3 C), the SEI layer resistance had more impact (2% voltage drop) on voltage profiles.

In future studies, this model will be further extended by including additional side reactions such as lithium plating. One key challenge in fast charging is lithium plating and its interaction with other side reactions. This means fast charging needs an adaptive charging protocol as a function of cycle. This model will be a very important foundation for the creation of a fast charging protocol.

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